Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/22/2015
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Project Title:	Epigenetic and Protein Expression Pattern Profiling of Caenorhabditis elegans Exposed to Time-Varying Gravitational Fields: a Multi-Generational Study		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AM08G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We hypothesize that long term temporal changes in the gravity vector will result in animal adaptation via epigenetic mechanisms that are passed onto subsequent generations. We further hypothesize that the gravity-influenced epigenetic profile can be restored to the consensus wild-type (i.e., Earth born) upon introduction to stable 1xg conditions. We propose to utilize C. elegans as a model organism to address this hypothesis by performing the following experiments: 1) Determine C. elegans behavioral responses (i.e., changes in migration dynamics, animal body posture, metabolism, etc.) due to temporal changes in the gravity vector. The gravity vector will be controlled by placing the animals inside a rotating CubeSat microscope developed by our group. The frequency, direction, and duration of rotation will be varied to elicit the most dramatic behavioral responses. Habituation time will be evaluated, and also changes in lifespan. 2) Identify gravity responsive genes by utilizing the vast collection of fluorescent strains of C. elegans available to researchers. We will monitor gene expression by fluorescence microscopy--from the embryonic stage through adulthood--during changes in the gravity vector. Genes of interest include those associated with muscle development, cell apoptosis, and neuronal development. 3) Profile the epigenetic marks on C. elegans exposed to long periods of continuous gravity vector changes that range from 1/4 of an animal's lifespan to three generations. We will examine the epigenetic consequence of restoring constant 1xg conditions over successive generations. The three Specific Aims can be conducted with the same animal strains, and thus allows the opportunity to correlate the findings across different experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding gravity's affect on biological systems is not well studied because this parameter is difficult to manipulate during experiments. The International Space Station houses the newest U.S. national laboratory, which can be utilized to understand the influence of microgravity on molecular, cellular, and animal model systems. Determining changes in epigenetic profiles during multi-generations exposed to different environments is important to know to prepare for future manned spaceflight beyond Earth. Additionally, temporary alteration of epigenomes is relevant to human health, and similar adaptation mechanisms may be displayed by almost any change in the environment or diet. In order to expedite molecular and cellular research in microgravity, we developed an innovative clinorotation microscope that enables us to observe living organisms, in real-time, during microgravity. Because of the extended clinorotation times (4 days to 3 months), we also developed a novel automated "worm chip" that permits hands-free feeding and maintenance of the animals, without interrupting clinorotation.
Task Progress:	We have made significant progress with our genetic and microscopy experimental procedures, and have a solid bioinformatics pipeline in place. Several highlights are presented in this report. * C. elegans Habituation and Reproduction (CeHR) media produces a phenotype whereby L1 larvae are internally hatched, and the parent is still viable. CeHR is an axenic liquid culture media used in the past by the International C. elegans Experiment First-Flight Development (ICE-First) mission. It has been known that these liquid cultures yield animals with low brood size (about 1/10th compared to agar-grown animals); however, the cellular and molecular mechanisms that cause these deformities are unknown. Using a DNA-specific fluorescent dye, we find that CeMM and CeHR liquid cultures produce animals with grossly deformed reproductive organs. We also tested liquid cultures with OP50 food source, and found that this culture does not produce the deformed reproductive organs as CeMM or CeHR, suggesting that these reproductive abnormalities are a result of the animals' diet, and not the liquid environment. Recently, we used ced-3 as a marker to identify apoptotic cells, and found that this gene is highly expressed in the germline in CeHR and CeMM animals, while there is little evidence of germline apoptosis in worms that are fed OP50. We hypothesize that the checkpoint for a conserved regulation of mitotic to meiotic cells in the germline is aberrant in CeHR and CeMM cultures. * C. elegans CeHR cultivation using an automated platform. We have cultured C. elegans for over four months, unattended, inside our novel "WormPharm" platform. This system could be utilized on the International Space Station (ISS) to perform long-term experiments with minimal crew intervention. * Bioinformatic analysis suggest that CeHR causes abnormalities in germline cells. Genes with known functions under the putative regulation of H3K37Me3 were correlated against each other based on their RNA expression profiles. Gene networks can be predicted based on the positive correlation of gene expression. Gene Ontology (GO) terms suggests an increase in genes associated with reproduction, namely vulva development. * The AB1 wild isolates of C. elegans adapt faster to liquid cultures compared to the N2 lab strain. The AB1 strain grows almost twice as fast in liquid cultures compared to N2. We hypothesize that "domestication" of the N2 strain has led to its decreased ability to adapt to a change in the environment (presumably due to loss of functions). This finding is important to consider when comparing genetic results of C. elegans with humans, and AB1 animals would be the preferred model for space flight experiments. * Histone marks exhibit "plastic" and "eleastic" memory. ChIP-Seq is performed with five histone marks: H3K27Me3, H4K20Me1, H3K4Me3, H3K9Me3, H3K36Me1, and RNA polymerase. The first generation (F0) of worms are grown in agar-filled petri dishes, followed by the second generation in liquid (F1), and "reverted" back to agar dishes for F2. It has been previously reported that H3K4Me3 modifications are broadly distributed, and we observe similar behavior for H3K27Me3. These broadly distributed marks are "plastic" in nature, in that their distributions are fairly constant when the environment changes. We defined the other marks as "elastic" because their distributions change in response to different environments. Surprisingly, an entirely new set of 279 genes is regulated by H3K27Me3 when animals are reverted from CeHR back to solid (agar) media; in contrast, H4K20Me1 only shows one new gene. Analysis was performed on histone modifications in the gene promoter regions only. The functions of a majority of these newly identified genes are currently unknown. We are performing gene silencing assays using RNA interference to understand their functions. * Histone modification H3K27Me3 exhibits plastic-like memory. Hierarchical analysis of different histone marks in different environments shows H3K27Me3 clustering. The other histone marks show clustering based on the environment, suggesting a more adaptive, elastic-like behavior. * Multiple histone marks can regulate a single gene. F0 Animals grown in agar show predominantly one histone modification in the promoter region of genes. F1 progeny grown in either liquid CeHR or OP50 contain multiple epigenetic marks for each gene promoter. CeHR F2 reversion animals display a shift from multiple marks per promoter, to one mark per promoter. However, OP50 F2 reversion animals still maintain multiple marks per promoter, which may

	provide a genetic basis for their inability to span multiple generations in OP50 cultures. * At present, we find no obvious correlation between the different histone marks and RNA transcription profiles obtained from RNA sequencing. However, we have not looked at all possible histone modifications, or correlations between RNA profiles and multiple histone marks. It may be the case that another form of regulation, for example small RNAs, are under the control of histone marks, and these small RNAs in turn regulate specific gene expression. * We are now comparing our ground-based simulated microgravity RNA transcription profiles with published space flight data (Florian Selcha, Akira Higashibata, Mari Imamizo-Sato, Atsushi Higashitani, Noriaki Ishioka, Nathaniel J. Szewczyk, and Catharine A. Conley. Genomic response of the nematode Caenorhabditis elegans to spaceflight. Adv Space Res. 2008; 41(5): 807–815.) In brief, our simulated microgravity RNA transcription profiling shows ~1300 genes are upregulated 4-fold (~5200 2-fold), and 68 downregulated 4-fold (671 2-fold) compared to controls. Selcha et al. reported 19 genes were upregulated and 10 genes were downregulated in response to space flight (based on 4-fold cutoff). All of the genes upregulated in response to space flight are also upregulated in our ground-based simulated microgravity experiments. Based on 2-fold cutoff, 10 of these genes are significantly upregulated, and five of these genes stay significantly upregulated when we use 4-fold as cutoff. Of all the 10 genes that were downregulated in response to space flight, we find eight of the genes similarly downregulated (out of 68 identified) in our simulated microgravity. Thus, our RNA profiling data suggests that clinorotation simulated microgravity of C. elegans is an appropriate analog for microgravity. This finding gives confidence in designing epigenetic assays for on-orbit studies based on clinorotation data.

Bibliography Type:	Description: (Last Updated: 08/17/2023)
Abstracts for Journals and Proceedings	Doh JH, Jayasinghe AK, Moore AB, Zelaya D, McCullough M, Moore MT, Sabanayagam CR. "Development of a Clinorotation Fluorescence Microscope to Study Cellular Processes in Live Microorganisms." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. Concurrent Session 11: Invertebrate Model. , Oct-2014
Abstracts for Journals and Proceedings	Doh J, Celen I, Sabanayagam C. "Revisiting the C. elegans maintenance media (CeMM) for automated, multi-generational cultivation, genome-wide analyses and pharmaceutical studies." Presented at the 2015 American Genetics Society Annual C. elegans Meeting, University of California Los Angeles, June 24-28, 2015. 2015 American Genetics Society Annual C. elegans Meeting, University of California Los Angeles, June 24-28, 2015. Program Number: 1052A. , Jun-2015
Abstracts for Journals and Proceedings	Celen I, Doh J, Sabanayagam C. "Epigenetic memory in C. elegans: Plastic and elastic behaviors of histone modifications in response to rapid environmental changes." Presented at the 2015 American Genetics Society Annual C. elegans Meeting, University of California Los Angeles, June 24-28, 2015. 2015 American Genetics Society Annual C. elegans Meeting, University of California Los Angeles, June 24-28, 2015. Program Number: 19. Search program: http://www.genetics-gsa.org/celegans/2015/pdf/2015%20Abstract%20Book.pdf ; accessed 10/27/15. , Jun-2015
Abstracts for Journals and Proceedings	Sabanayagam CR, Jayasinghe AK, Doh JH, Celen I, Moore A, Moore MT. "Worms in Space: Epigenetic Response of C. elegans in Simulated Microgravity." Presented at the 59th Biophysical Society Meeting, Baltimore, MD, February 7-11, 2015. Biophysical Journal. 2015 Jan;108(2, Suppl 1):464a. http://dx.doi.org/10.1016/j.bpj.2014.11.2532 , Feb-2015
Significant Media Coverage	Skirble R. (Interview with CR Sabanayagam.) "Voice of America video interview: Experiments on Worms May Keep Astronauts Healthier in Space." Voice of America video interview, February 12, 2015. http://www.voanews.com/content/experiments-worms-astronauts-healthier-space/2641127.html ; accessed 10/27/15., Feb-2015
Significant Media Coverage	Rogers K. (Interview with Sabanayagam CR) "Motherboard interview: See-Through Space Worms May Hold the Key to Living on Mars." Motherboard interview, February 13, 2015. http://motherboard.vice.com/read/see-through-space-worms-may-hold-the-key-to-living-on-mars ; accessed 10/27/15., Feb-2015
Significant Media Coverage	Sabanayagam CR. "Space Worms." BBC, The Naked Scientist interview, February 15, 2015. Can download mp3; see http://www.thenakedscientists.com/HTML/specials/show/20150215/ ; accessed 10/27/15., Feb-2015